

WORLD MODELS • ARXIV 2509.24527

Dreamer 4

A world model good enough
to train an agent in

Training Agents Inside of Scalable World Models

Hafner*, Yan*, Lillicrap (Google DeepMind), September 2025

Presenter _____ Date _____ ~30 min + discussion

Right, all generated by the Dreamer 4 world model: an imagined rollout, a human playing inside the model, and the robot-arm model.



The paper in three minutes: the official video

'Dreamer 4 | Diamonds from Offline Experience', Danijar Hafner's channel, 1 Oct 2025, 2:55. Figure 1 in motion



► Click the picture to open the video on YouTube (internet required): youtube.com/watch?v=oDIBtTcX0g0

Project page: 60 more clips (imagination rollouts, 17 interaction tasks × 3 models, robot arm): danijar.com/project/dreamer4



Scan to watch on your own device

2:55 · uploaded 1 Oct 2025

YouTube channel @danijar (Danijar Hafner)

Description: 'the first agent to mine diamonds learning entirely offline'

WHAT TO WATCH FOR

The imagined training rollouts of Figure 1 in motion (chopping, crafting menus, mining, cave lighting, diamond ore), decoded from latents for viewing

The headline it builds up to: diamonds from 2,541 h of offline video, no environment interaction

NO INTERNET? EMBEDDED 10-S TEASER, CLICK TO PLAY



The thesis in one slide

Three statements and three numbers

- 01 World models became media.** In 2024–25, Genie, Oasis, Lucid-v1 and GameNGen showed real-time, playable video worlds. They were judged by how they look and feel; whether an agent could learn in them was not asked.
- 02 An agent needs more than looks.** To train in a model, the mechanics must be right (menus, crafting, tools), it must run fast enough for millions of rollouts, and it must stay stable over long horizons. The playable models failed the first test; the agent-side models (Dreamer 1–3) passed it only at 64×64 pixels with privileged state.
- 03 Dreamer 4 is the first to do both.** Humans solve 14 of 16 interaction tasks inside it in real time, and an agent trained only inside it, from 2,541 h of offline video, reaches diamonds. No environment interaction at any point.

THREE NUMBERS TO REMEMBER

14 / 16: interaction tasks human testers solve inside the model, live, at 21 FPS (Table 1; best prior Minecraft model: 5 / 16)

21 FPS on one H100, with a 9.6 s context and four sampling steps per frame; the same design at 64 steps ran at 0.8 FPS (Table 2)

29 % · 0.7 %: iron pickaxe and diamond success from offline video only; a Gemma-3 VLA on the same data: 11 % · 0 %, plain BC: 0.6 % · 0 % (Table 7)

Route: Part 1, what a world model is and where the field stood; Part 2, six things Dreamer 4 does differently; Part 3, the evidence; Part 4, where it stands and what to take away.

PART 1

World models

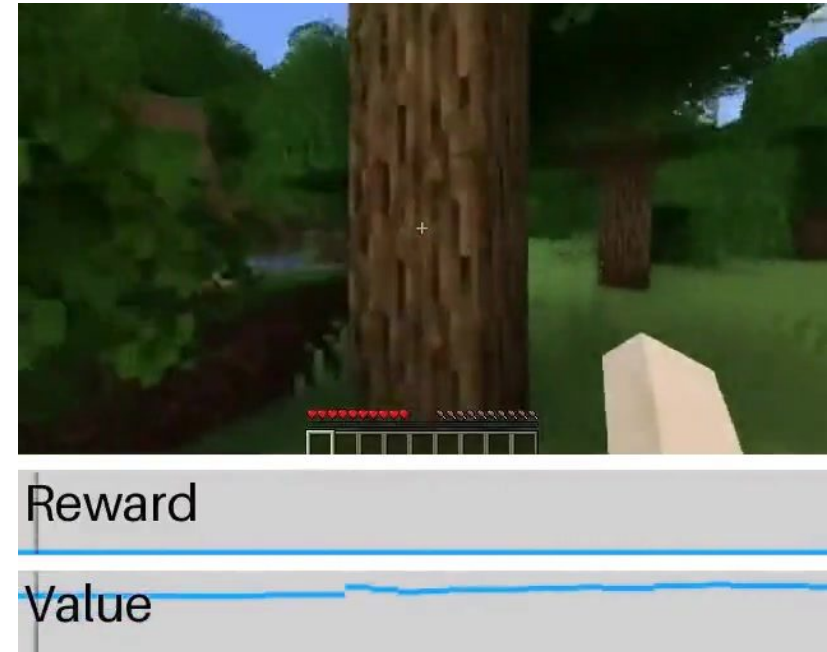
What they are, what the 2024–25 wave delivered, and what an agent needs from one

01

What a world model is, and what 'learning in imagination' means

The Dreamer lineage: same idea, increasingly capable simulators

- A world model is a learned simulator: given the current situation and an action, it predicts what happens next
- Imagination training optimises the policy on rollouts generated *by the model* instead of by the real environment, which makes experience cheap, parallel and safe
- Why it matters now: robots and other real-world systems cannot afford millions of trial-and-error interactions, while video of the world is abundant
- Dreamer 4's bet: if the simulator gets object interactions right, RL inside it can go beyond what pure imitation learns



Imagined rollout decoded for visualisation, with the agent's reward and value estimates (project page, 'Gather wood').

Dreamer (2019)

latent imagination on DM Control

DreamerV2 (2020)

discrete latents, Atari

DreamerV3 (2023/Nature 2025)

one config for 150+ tasks; diamonds with online RL

Dreamer 4 (2025)

transformer diffusion WM; diamonds offline; real-time on 1 GPU

Two families of world models, and where Dreamer 4 sits

2019–2025: models an agent trains inside vs. models a human plays inside (numbers from Table 1 and the vendors' announcements)



- Two yardsticks: agent-side models are judged by the agent's return, media-side models by how the video looks. Dreamer 4 is held to both, plus a third: does an agent trained *only* inside it succeed in the real game?
- The direct baselines are the playable Minecraft models (Table 1); Genie 3 is the closest in ambition but exposes no comparable action space, so the paper cannot compare against it (§4.2)

What an agent needs from a world model, and why that was hard

Three requirements that pulled in different directions

01

Fidelity of interactions

Video generators look good but get game mechanics wrong: inventories, crafting menus, breaking blocks, placing a boat and riding it.

Prior Minecraft world models (Oasis, Lucid-v1, MineWorld) fail most such tasks.

02

Speed

Imagination RL needs millions of generated frames, and a human tester needs at least 20 FPS.

Diffusion transformers spend tens of denoising steps per frame: 0.8 FPS for the baseline in this paper.

03

Memory & drift

Autoregressive video drifts: small errors feed back into the context.

Short contexts (1–2 s in prior models) forget what is behind the player.

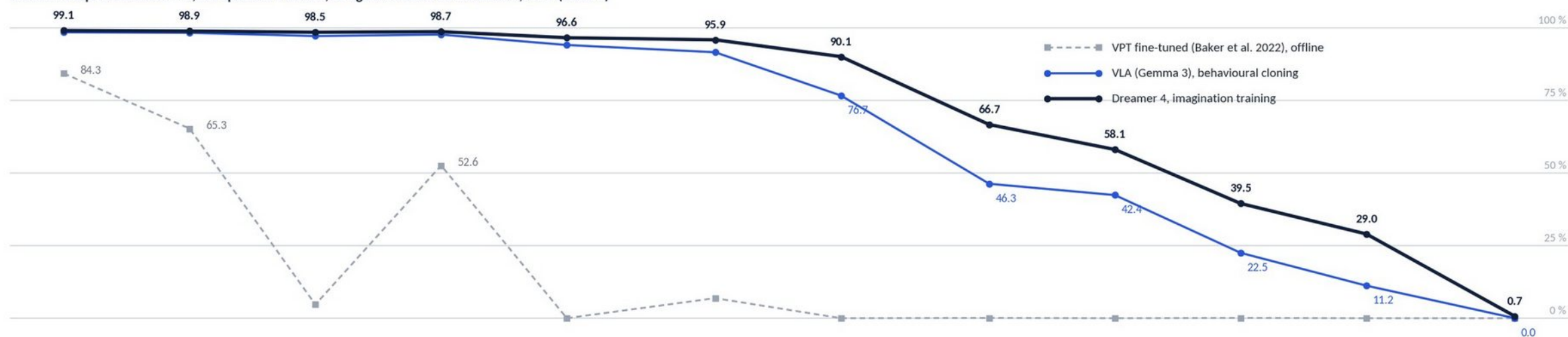
- Dreamer 3's RSSM is fast (about 1,000× faster than a diffusion transformer) but works at 64×64 pixels and needed abstract inventory state and crafting actions instead of raw mouse and keyboard input
- Dreamer 4's answer: a diffusion transformer that needs only 4 sampling steps (shortcut forcing), an architecture tuned for long context at low cost, and training choices that keep long rollouts stable (x-prediction, noised context)

The test: twelve milestones, three tool gates, ~24,000 actions

2,541 h of human gameplay video, no environment interaction; where each agent falls off (Tables 7 and 8)



Success rate per milestone — 1,000 episodes × 60 min, all agents trained on the same 2,541 h (Table 7)



Human players with Minecraft experience need about 20 minutes ≈ 24,000 mouse-and-keyboard actions to reach a diamond; an evaluation episode gives the agent 60 minutes = 72,000 actions. The chain cannot be shortcut: each item needs the previous tool, and mining a block with a weaker tool destroys it without a drop. Diamonds sit at the end of twelve dependent skills — crafting menus, mouse precision, digging, smelting — all learned from offline video.

PART 2

What makes Dreamer 4 special

Six design decisions: latents, shortcut forcing, a fast architecture, the agent inside, offline RL, cheap labels

02

What makes Dreamer 4 special: six design decisions

Each maps to a slide in this part; the phase of the recipe it belongs to is in brackets

- 01 Latents, not pixels.** A causal tokenizer compresses each frame to 256 continuous tokens; simulation and RL happen in latent space, pixels are only decoded for people [phase 1]
- 02 Shortcut forcing: diffusion quality in four steps.** Diffusion forcing + shortcut models, with x-prediction and an x-space loss, so long rollouts stay sharp and fast [phase 1]
- 03 An architecture tuned for 21 FPS on one GPU.** Space attention in most layers, causal time attention every 4th, grouped-query attention, registers; 9.6 s of context in real time [phase 1]
- 04 The agent lives inside the world model.** Agent tokens with one-way attention read the simulator's state; policy, reward and value are small heads on the same transformer [phase 2]
- 05 Imagination training, offline, with PMPO.** The transformer is frozen; policy and value heads learn from imagined rollouts scored by the learned reward head, anchored to the BC prior [phase 3]
- 06 Action semantics from few labels.** Unlabeled video gets a learned 'no action' embedding; ~100 h of labels suffice and the actions transfer to worlds seen only unlabeled [phase 1]

THE RECIPE (ALGORITHM 1)

Phase 1 • World-model pretraining: tokenizer (masked autoencoding), then the dynamics model with shortcut forcing on all 2,541 h, labeled or not

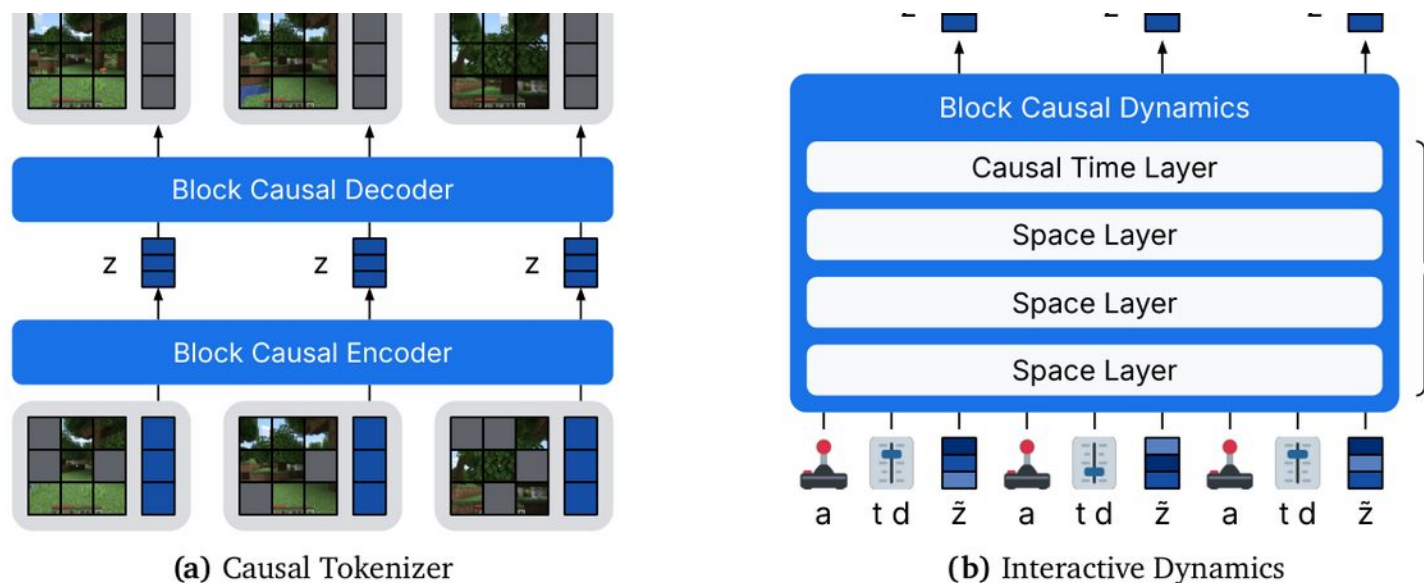
Phase 2 • Agent fine-tuning: add agent tokens, policy and reward heads; behavioural cloning with multi-token prediction while the dynamics loss keeps running

Phase 3 • Imagination training: freeze the transformer; train policy and value heads by RL on rollouts inside the model

One block-causal transformer throughout; the agent is a few heads on top of the simulator.

1 • One transformer, two roles: causal tokenizer + dynamics model

Figure 2: causal tokenizer + interactive dynamics model, one shared block-causal transformer

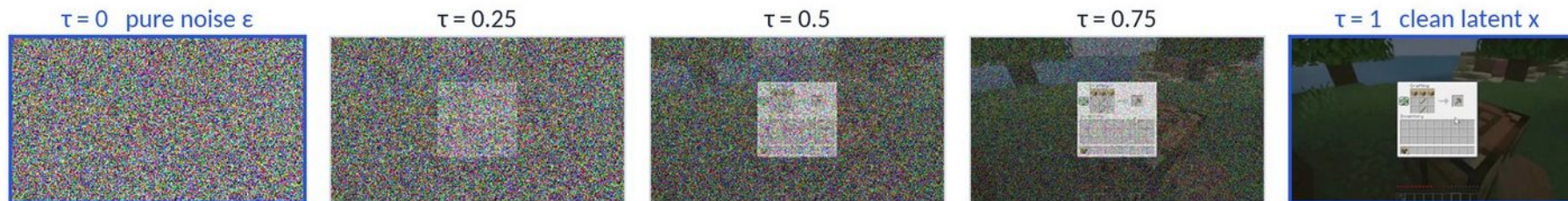


Left: causal tokenizer (masked patches $\rightarrow$ latent tokens $\rightarrow$ reconstruction). Right: interactive dynamics with interleaved action tokens a , latent tokens z , and per-frame signal level τ / step size d ; space layers with a causal time layer every 4th block. (Figure 2, arXiv 2509.24527)

- **Parameters:** 2 B in total, $\approx$ 400 M tokenizer + 1.6 B dynamics
- **Frames:** 360 $\times$ 640, zero-padded to 384 $\times$ 640, i.e. 960 patches of 16 $\times$ 16
- **Latents:** $N_z = 256$ tokens $\times$ 32 dims per frame (512 $\times$ 16 bottleneck, tanh)
- **Context:** 192 frames = 9.6 s at 20 FPS
- **Attention:** block-causal, full within a frame and causal across frames, so the model runs online frame by frame
- **Block recipe:** pre-RMSNorm, RoPE, SwiGLU, QK-norm, logit soft-capping, register tokens

2 • Primer 1 of 3 • Flow matching: one dial from noise to image

The background the dynamics objective is built on (paper §2)



sampling: start at $\tau = 0$ and walk to $\tau = 1$ in K steps of size $d = 1/K$, $x(\tau+d) = x(\tau) + f(x(\tau), \tau) \cdot d$

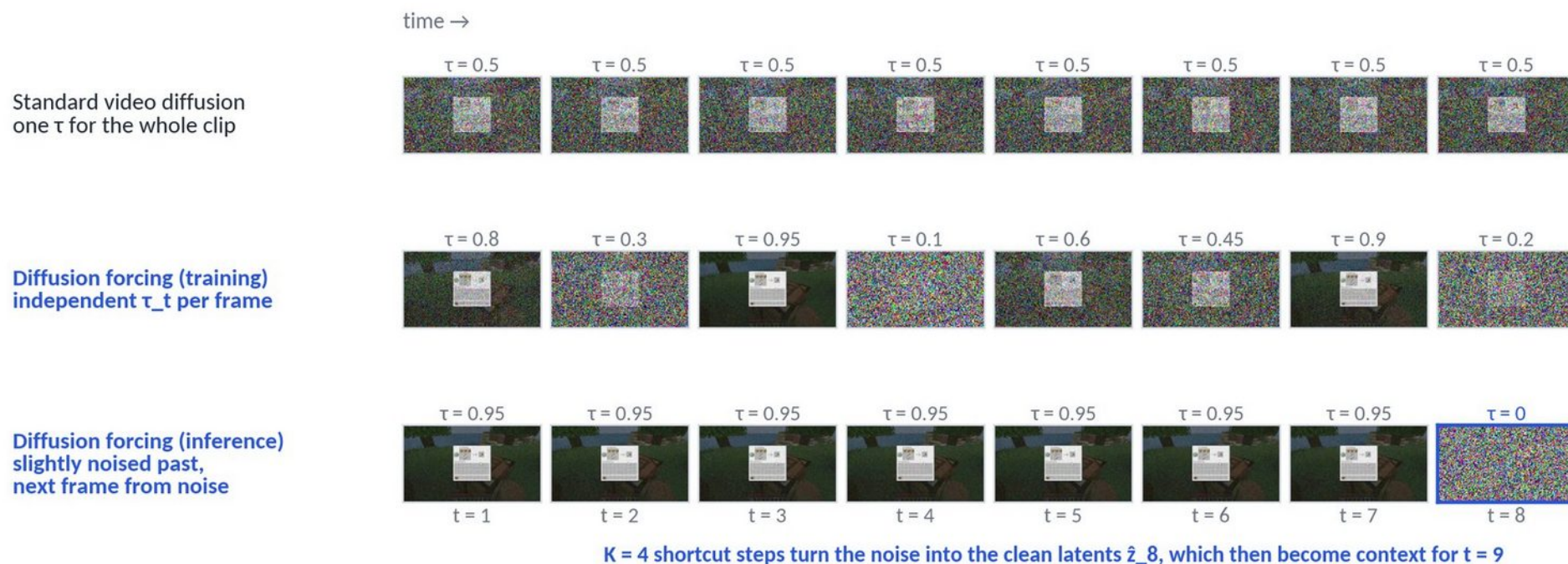
training: pick a random τ , corrupt the clean latent $x(\tau) = (1 - \tau) \cdot \epsilon + \tau \cdot x$, and ask the network for the way back —

either the velocity $v = x - \epsilon$ (v-prediction) or the clean latent x itself (x-prediction); they carry the same information: $v = (x - x(\tau)) / (1 - \tau)$

- **Signal level $\tau \in [0, 1]$:** $x(\tau) = (1 - \tau) \cdot \epsilon + \tau \cdot x$, where $\tau = 0$ is pure noise and $\tau = 1$ the clean latent; during training τ is drawn at random for every example
- The network learns a direction: $f(x(\tau), \tau) \approx v = x - \epsilon$, the velocity pointing from noise to data; sampling starts from noise and walks to $\tau = 1$ in K steps
- Two equivalent parameterisations: predict the velocity v (standard) or the clean latent x directly (x-prediction); the maths is identical, but the error behaviour under autoregressive feedback differs (see the x-prediction slide)
- Why so many steps? The learned direction is only locally right, so ordinary diffusion needs 16–64 small steps per image, far too slow for a 20 FPS simulator

2 • Primer 2 of 3 • Diffusion forcing: a noise level per frame

Chen et al., 2024: what makes causal, frame-by-frame generation possible



- Standard video diffusion corrupts the whole clip with one τ : past and future are always equally noisy, so the model cannot be rolled out one frame at a time
- Diffusion forcing draws an independent τ_t for every frame; each frame is at once a denoising target and (noisy) history for the frames after it, with a single loss over the whole sequence
- What it buys at inference: the next frame is generated from pure noise given a clean or lightly noised past, which is exactly the interactive setting; keeping the past slightly noised makes the model tolerant of its own small mistakes
- What it does not fix: still tens of denoising steps for every frame, the 0.8 FPS starting point of the design cascade (Table 2)

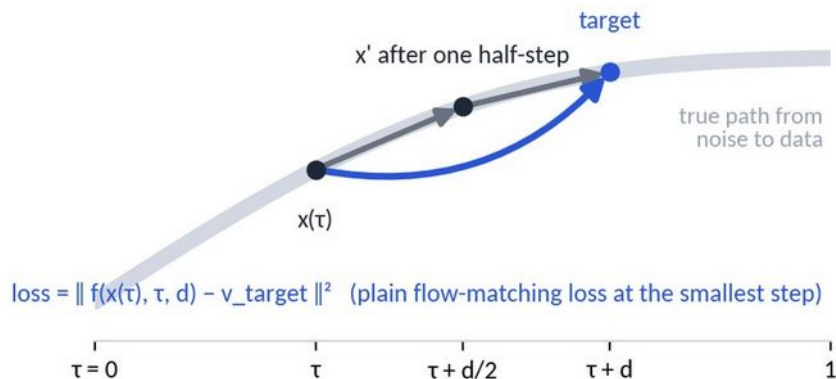
2 • Primer 3 of 3 • Shortcut models: learning to take big steps

Frans, Hafner, Levine, Abbeel, 2024: why four sampling steps can be enough

Training: step-size conditioning + self-consistency

grey: two half-steps of size $d/2$, computed by the model itself (stop-gradient) $\rightarrow$ target

blue: one step of size d must land at the same point



Sampling one frame

shortcut model: $K = 4$ steps of $d = 1/4 \rightarrow 21$ FPS



ordinary diffusion / diffusion forcing: $K = 64$ small steps $\rightarrow 0.8$ FPS



same network, same training run — no separate distillation stage

- Add the step size d as an input: $f(x(\tau), \tau, d)$ predicts the average direction over a whole jump of size d instead of only the local direction
- Smallest step: ordinary flow-matching loss. Larger steps: a self-consistency (bootstrap) loss; one step of size d must land where two steps of size $d/2$ land, and the two-step target is produced by the model itself with gradients stopped
- Result: 2–4 steps reach the quality of 64 in the original paper, with no separate distillation stage (one model, one training run)
- Dreamer 4 combines the two: per-frame τ_t from diffusion forcing plus step-size conditioning from shortcut models gives shortcut forcing, $K = 4$ steps per frame at 21 FPS (next slide)

2 • Shortcut forcing: the training objective

Diffusion forcing + shortcut models, with three engineering choices that turn out to matter a lot

- **Per-frame noise:** each frame gets an independent signal level τ_t ; the model predicts frame t 's clean latent from the noisy frame t , past frames, and actions
- **Step-size conditioning:** the network also receives d ; for $d > d_{\min}$ the target is the *bootstrap* target: the result of two half-steps computed by the model itself (no gradient through the target)
- **Choice 1, x-prediction:** the network outputs the clean latent x (v-prediction would output the velocity)
- **Choice 2, loss in x-space:** the bootstrap target is formed in v -space (where two half-steps compose cleanly) but the loss is scaled back to x -space by $(1 - \tau)^2$
- **Choice 3, ramp weight:** the loss weight $w(\tau) = 0.9\tau + 0.1$ favours high-signal (low-noise) levels, where fine detail is decided
- **Inference:** $K = 4$ steps per frame; context frames at $\tau_{\text{ctx}} \approx 0.1$; sampling stays stochastic, so RL sees diverse imagined futures

EFFECT OF EACH PIECE (TABLE 2, FVD ↓)

Diffusion forcing, $K = 64$ steps: 306 (0.8 FPS)

Same model, $K = 4$ steps: 875

+ shortcut objective: 329

+ x-prediction: 326

+ x-space loss: 151

+ ramp weight: 102

Full architecture but v -space prediction/loss: 124 vs 57

WHY THIS MATTERS MOST

4 steps $\approx$ the quality of 16–32 diffusion-forcing steps (Fig. 8), roughly 16 $\times$ fewer network evaluations per frame. That is what makes real-time interaction and large-scale imagination RL feasible.

2 • Why x-prediction matters for autoregressive world models

Hafner's explanation (TalkRL, Nov 2025) and the numbers

- v-prediction carries the noise: to output $v = x - \epsilon$ the network must reproduce the *exact* noise pattern of its input through every layer; small mistakes in that noise become errors in the sample
- In autoregressive use these errors compound: each generated frame becomes context for the next, and velocity errors accumulate as drift, blur or flicker over long rollouts
- x-prediction is a denoising target: the network only needs to know what the clean frame should look like, so errors are pulled back toward plausible images at every step
- Combined with noised context ($\tau_{\text{ctx}} \approx 0.1$), the model is trained and run on slightly corrupted context, so it never sees a distribution it was not trained on
- Evidence: the full architecture with v-space prediction and loss reaches FVD 124; with x-prediction and x-space loss, 57 (Table 2). FVD was computed on long 384-frame generations to expose compounding error

Generation quality for sampling steps

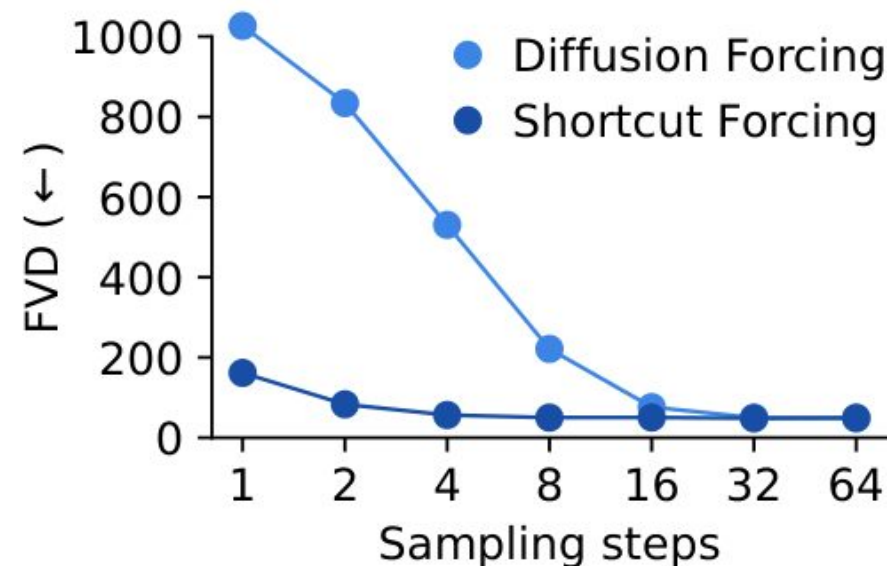
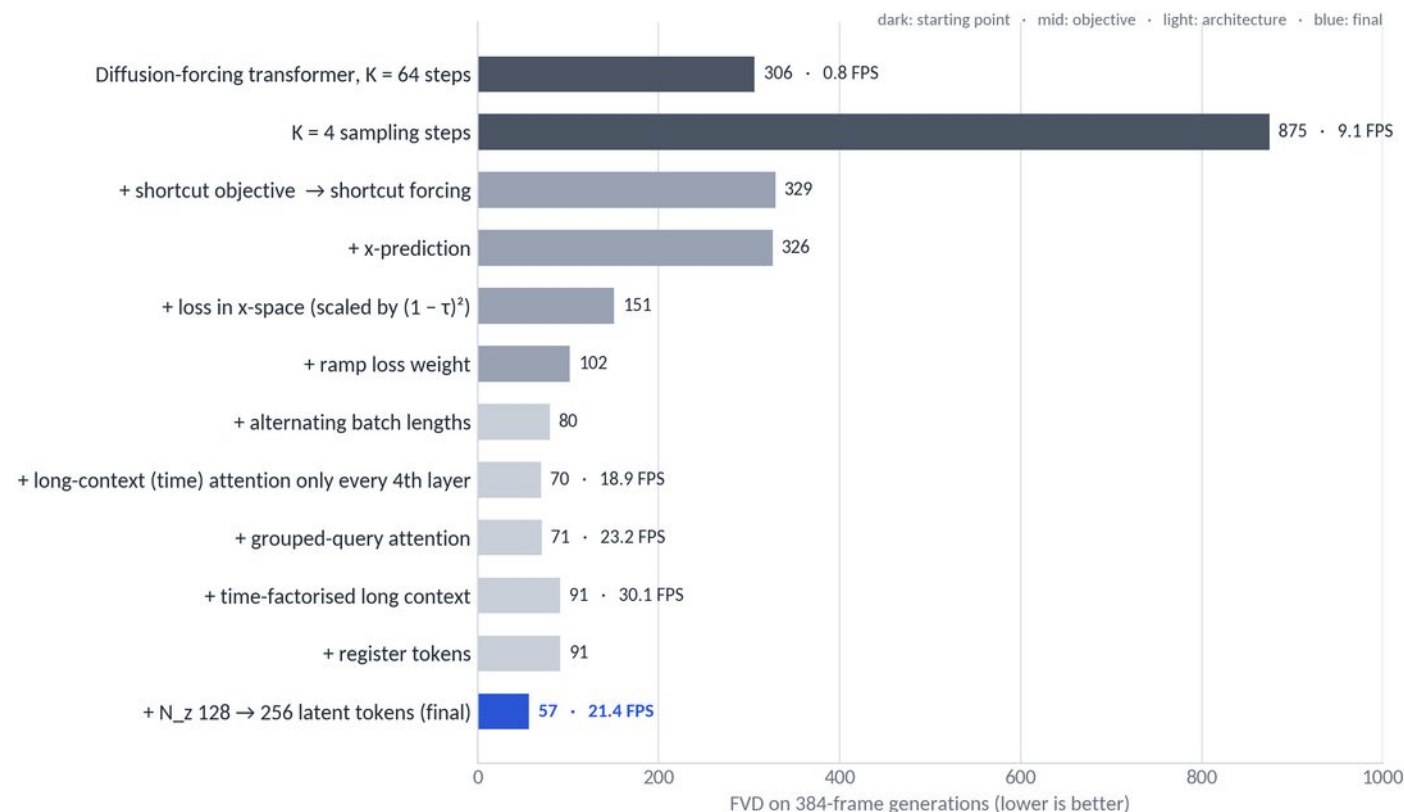


Figure 8: FVD vs sampling steps. Shortcut forcing at 4 steps ≈ 60 ; diffusion forcing needs 16–32 steps to match.

3 • Architecture choices that buy speed without losing quality

Table 2: each row adds one change to the previous row



- Alternating batch lengths (64 / 256 frames): FVD 102 → 80 and a training step from 9.8 s to 1.5 s
- Time attention only every 4th layer: 18.9 FPS at FVD 70
- Grouped-query attention: 23.2 FPS, FVD unchanged
- Time-factorised long context trades a little quality (91) for 30 FPS; registers stabilise attention
- More latent tokens (128 → 256) bring FVD to 57 at 21.4 FPS, the final configuration
- Runs were 48 h each; FVD computed on 1,024 generations of 384 frames

4 • From world model to agent: agent tokens & heads

Phase 2: policy and reward live inside the same transformer but cannot disturb the simulation

- **Agent tokens:** extra tokens per frame that carry a task embedding (one-hot over the 20 training tasks) and read out policy and reward
- **One-way attention:** agent tokens attend to actions, latents and past frames, but *nothing attends back to them*, so the world model never conditions on the policy's intentions or reward estimates
- **Policy head:** behavioural cloning with multi-token prediction of the next $L = 8$ actions; the extra horizon regularises the representation and encodes intent
- **Reward head:** predicts task rewards derived from game events (item obtained); a task-conditioned *scalar* reward was chosen over a reward vector so that open-vocabulary tasks remain possible
- **Value head:** added in Phase 3, trained by TD learning on imagined rollouts
- Result already at this stage: 'WM + BC' beats BC on every hard milestone (iron pickaxe 16.9 % vs 0.6 %), so the video-prediction representation is a strong prior for control

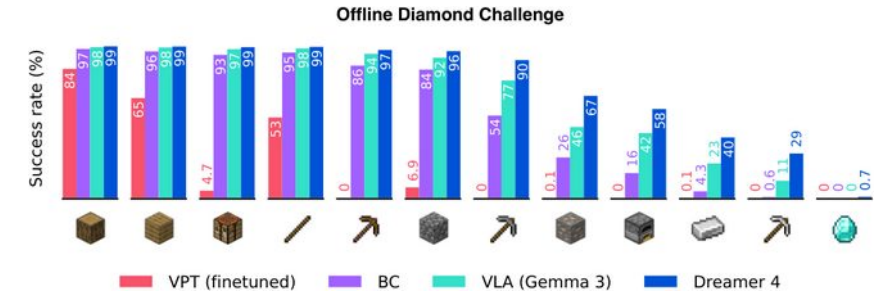


Figure 3: success rates of agents trained on the same offline data (BC, VLA, WM+BC, Dreamer 4).

WHY NOT A SEPARATE POLICY NETWORK?

Sharing the backbone gives the policy the world model's representation for free, and keeps everything one model. The one-way attention mask is what makes this safe.

5 • Imagination training with PMPO

Phase 3: RL on imagined rollouts, with the transformer frozen

- Rollouts: start from real context frames, let the policy act, and let the world model generate what happens next; rewards come from the learned reward head; $\gamma = 0.997$
- Frozen transformer: only the policy and value heads are trained, so the simulator cannot drift toward the policy
- PMPO (Abdolmaleki et al., 2024): instead of scaling the policy gradient by the advantage, use only its *sign*: positive-advantage actions are pushed up and negative ones down, with balanced weight $\alpha = 0.5$ on each set
- Reverse KL to the frozen BC policy with $\beta = 0.3$ keeps the policy on the data manifold, like KL-regularised RLHF; it is also the main guard against exploiting world-model gaps
- Why not Dreamer 3's recipe? The sign-based update replaces return normalisation and the entropy bonus; it is insensitive to reward scale and copes with the extremely sparse late-game rewards
- Effect: iron pickaxe 16.9 % $\rightarrow$ 29.0 %, iron ingot 27.8 % $\rightarrow$ 39.5 %, diamond 0 $\rightarrow$ 0.7 %; agents also become faster (Table 8)



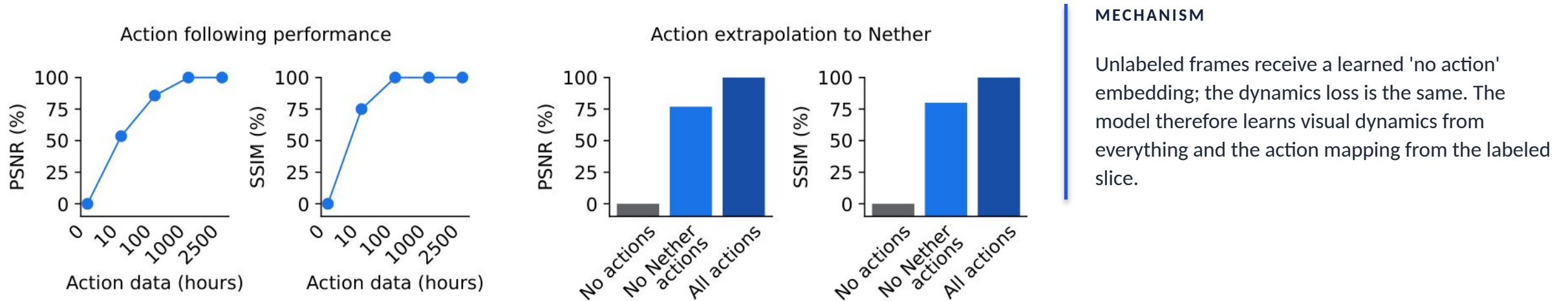
Imagined 'mine stone' rollout with reward and value traces (project page).

FAILURE MODE THE KL TERM ADDRESSES

Hafner's example: the policy discovers it can 'craft' a pickaxe from invalid materials because the model renders it anyway. The KL to the BC prior limits such exploitation.

6 • Unlabeled video: how many action labels are needed?

Figure 7: action conditioning from a small labeled fraction, and transfer to unseen visual domains



- **Setup:** train on all 2,541 h of video, but keep actions for only a subset; measure PSNR/SSIM of action-conditioned predictions, normalised between 'no actions' (0 %) and 'all actions' (100 %)
- 10 h of labels → 53 % PSNR / 75 % SSIM; 100 h → 85 % / 100 %; 1,000 h → 100 %. Action understanding saturates with roughly 4 % of the data labeled
- **Domain transfer:** with actions only from the Overworld, predictions in the Nether/End (seen unlabeled) recover 76 % / 80 %, so the model separates 'what actions do' from 'what the world looks like'
- Why it matters: this is the road to internet-scale video pretraining, where labels are scarce (VPT needed an inverse-dynamics model for this)

Does it hold up?

Humans inside the model, diamonds from offline data, and where the gains come from

03

Is the world model accurate? Let humans play inside it

Table 1: real-time interaction on one H100; 16 tasks attempted by human testers from the same starting frame

Model	Params	Resolution	Context	FPS	Tasks solved
MineWorld	1.2 B	384×224	0.8 s	2	not evaluable (needs actions in advance)
Lucid-v1	1.1 B	640×360	1.0 s	44	0 / 16
Oasis (small)	500 M	640×360	1.6 s	20	0 / 16
Oasis (large)	—	360×360	1.6 s	~5	5 / 16
Dreamer 4	2 B	640×360	9.6 s	21	14 / 16

- **The 16 tasks:** eat apple · place 3 torches · chop tree · dig 3×3 pit · craft wooden pickaxe · use furnace · place & open workbench · kill zombies · mine diamonds · complete window · place & ride boat · enter portal · place door · bed & sleep · plant reed & pour water · turn 360° & enter house
- The two failures were memory-related: the world changes when you look away for longer than the 9.6 s context, and inventory contents become unclear over time
- Genie 3 is not compared: it has no fine-grained mouse/keyboard control

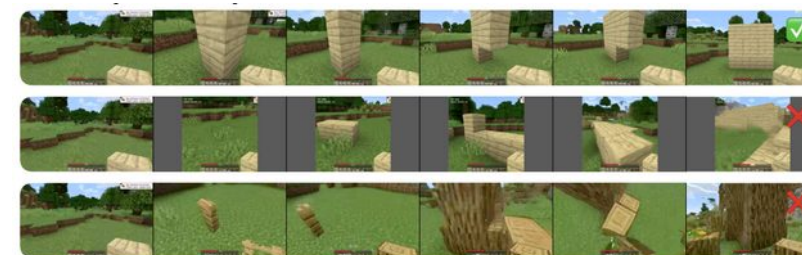


Figure 5: same task, same start frame. Dreamer 4 (top) builds the wall; Lucid-v1 and Oasis do not.

WHY THIS EVALUATION

FVD and PSNR measure how video *looks*; only interaction tests whether the model got the *mechanics* right. The paper complements it with FVD on 384-frame rollouts, but there is no long-horizon *quantitative* interaction metric yet, which is a fair critique.

What 'accurate object interactions' looks like

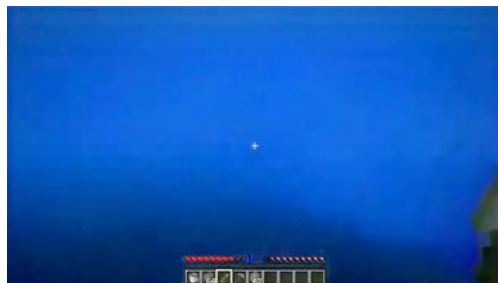
Frames from the project page's side-by-side human sessions (same start frame, same human inputs)

Dreamer 4

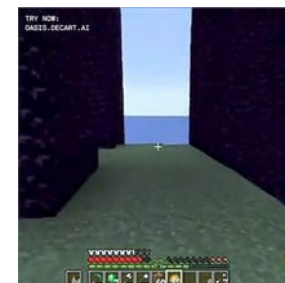
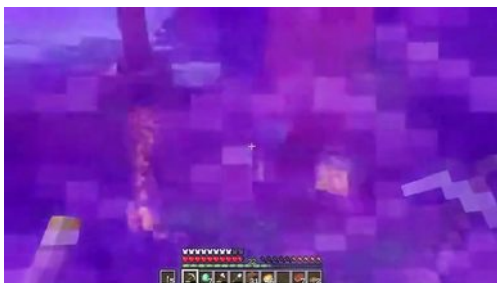
Lucid-v1

Oasis

Place and ride
boat



Enter portal



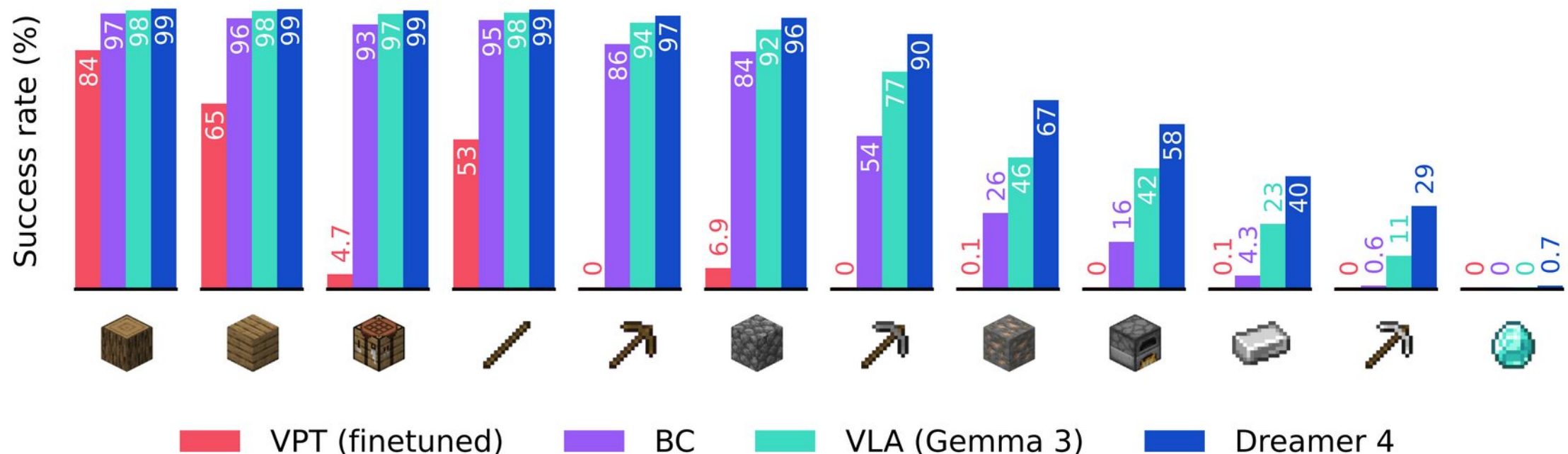
Craft wooden
pickaxe



Single frames sampled from the project-page clips. Dreamer 4 renders the boat ride, the Nether-portal swirl and a working crafting-table menu; the baselines lose the scene or the menu. ► All 51 clips (17 tasks × 3 models): danijar.com/project/dreamer4

Diamonds from offline experience: the results

Success rate per milestone; 1,000 episodes × 60 min; same 2,541 h of data for all methods (Table 7 / Fig. 3)

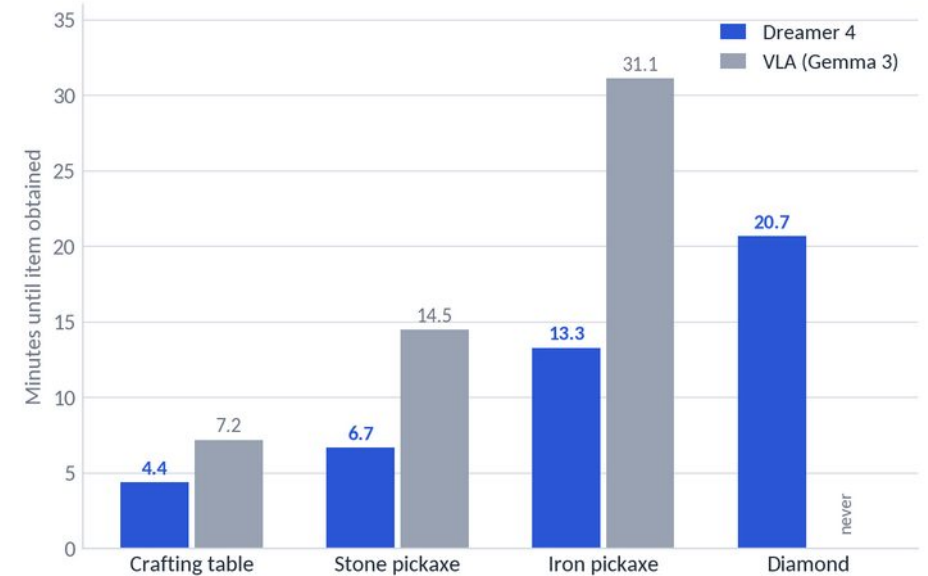
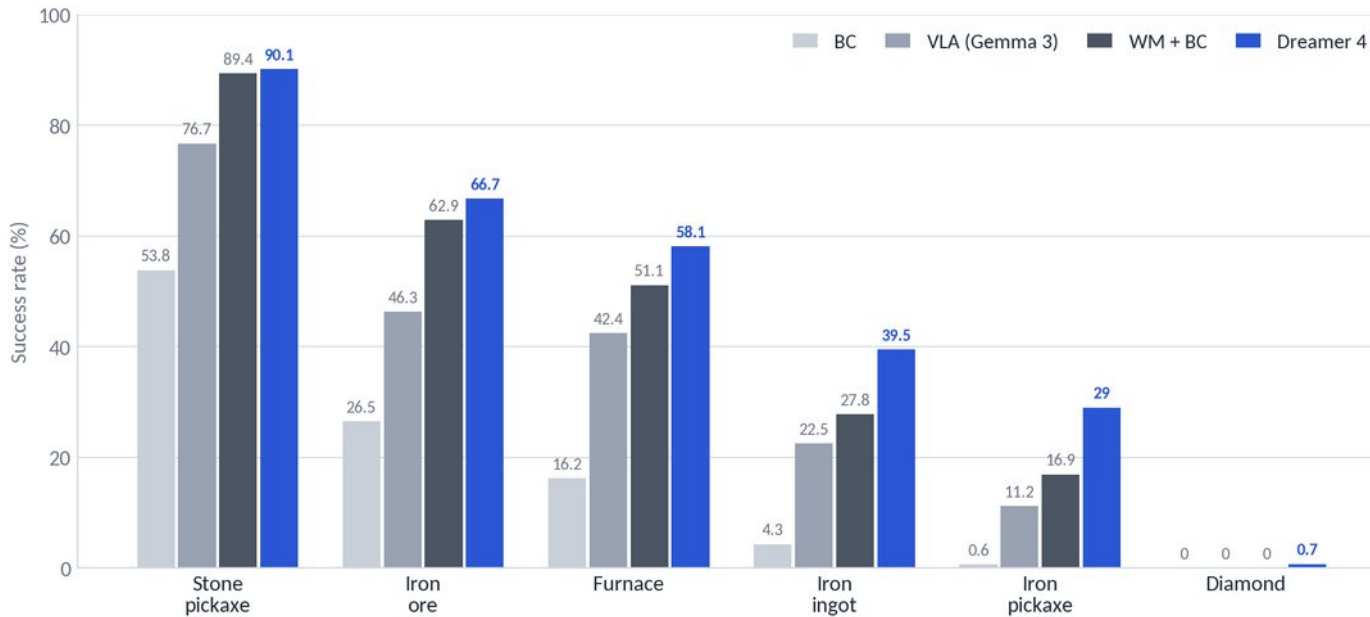


Official figure from danijar.com/project/dreamer4 (benchmark.png). ▶ Watch the trained agent in the real game, four uncut 60-minute evaluation episodes on YouTube: [Uncut #1](#) · [#2](#) · [#3](#) · [#4](#)

- **Dreamer 4:** stone pickaxe 90.1 % · iron ore 66.7 % · furnace 58.1 % · iron ingot 39.5 % · iron pickaxe 29.0 % · **diamond 0.7 %** (≈ 7 of 1,000 episodes)
- **VLA on Gemma 3 (same data):** iron pickaxe 11.2 %, no diamonds · **BC:** 0.6 % · **VPT fine-tuned** (pretrained on 270 K h of web video): only reliably reaches sticks
- The gap widens along the tech tree: imitation is enough for early items; imagination RL matters where data is thin and horizons are long

Where do the gains come from? Agent ablations and speed

Figure 4 / Table 8: representation (WM+BC vs BC) and imagination RL (Dreamer 4 vs WM+BC) both contribute



- **Representation:** BC on the world model's features (WM+BC) vs plain BC: iron pickaxe 16.9 % vs 0.6 %; the authors: 'video prediction implicitly learns an understanding of the world that is useful for decision making'
- **Imagination RL:** adds +12 pts on iron pickaxe and +12 pts on iron ingot over WM+BC, and unlocks diamonds
- **Speed:** Dreamer 4 is fastest at every milestone: iron pickaxe in 13.3 min vs 31.1 min for the VLA; diamonds in 20.7 min on average when achieved (human average $\approx$ 20 min)
- Task input matters: BC without the task embedding collapses to 8.8 % stone pickaxe (Table 7)

PART 4

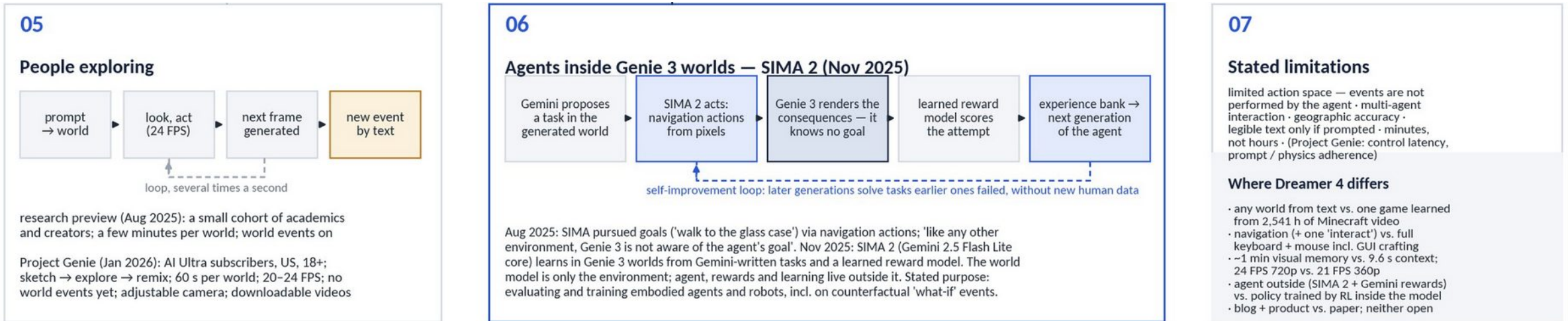
Where it stands

Genie 3, the caveats, and what to take away

04

Genie 3 and Dreamer 4: agent outside vs. agent inside the model

Human play in the research preview and Project Genie, the SIMA 2 agent loop, stated limitations



- Agent outside vs. agent inside: SIMA 2 acts in Genie 3 worlds with Gemini-written tasks and a learned reward model, and Genie 3 only renders the consequences; Dreamer 4's policy is a head on the same transformer, trained by RL on imagined latents with the reward head it learned from the data
- Same limits, different scale: both are closed. Genie 3 stays consistent for minutes but with a limited action space; Dreamer 4 forgets beyond 9.6 s but simulates crafting menus, furnaces, boats and portals from raw mouse and keyboard
- Convergent direction: both bet on world models as training grounds, Genie 3 via generated worlds for embodied agents and robots, Dreamer 4 via imagination training on offline video; neither has yet shown a policy that transfers to a physical robot

A critical reading: established vs. open

What a reviewer would push on (and what independent groups later found)

Well supported

- Shortcut forcing + x-prediction gives a large speed/quality win: a clean 12-step ablation cascade, reproduced by Open Dreamer at 1.6 B scale (JAX, 2026).
- The world model handles object interactions far better than Oasis / Lucid-v1: consistent human play-tests plus long-rollout FVD.
- World-model features are a strong prior for imitation (WM+BC >> BC on the same data).
- Action conditioning from ~4 % labeled data works and transfers across visual domains.

Open or thinly supported

- Diamond result: 0.7 % without confidence intervals or per-seed variance; success criterion is a game event, protocol details are in the appendix only.
- No direct quantitative long-horizon accuracy metric; interaction accuracy is human-judged on 16 tasks (Pith/EmergentMind critique).
- RL gains over WM+BC are modest on most items; small-scale reproductions report high seed variance in the RL phase and reward-hacking of the frozen reward head.
- No peer review as of Sept 2026 (arXiv v1 only); no code or weights; nobody outside DeepMind has reproduced offline diamonds.
- Generality of the *agent* beyond Minecraft is untested.

What makes Dreamer 4 special, in one slide

01

The first world model good enough to train in

Humans solve 14 of 16 interaction tasks inside it in real time; an agent trained only inside it reaches diamonds from offline video.

04

Imagination adds what imitation cannot

Most of the gain over BC is the world-model representation; RL inside the model adds the rest where data is thin: iron pickaxe 16.9 → 29 %, diamonds 0 → 0.7 %.

02

Real-time diffusion is the enabling trick

Shortcut forcing (4 steps), x-prediction with an x-space loss and time attention every 4th layer: 21 FPS on one GPU with 9.6 s of context.

05

Video is the prior, labels are cheap

About 100 h of action labels on top of thousands of unlabeled hours suffice; action semantics transfer to worlds seen only as video.

03

One transformer, three roles

Tokenizer, simulator and agent share the block-causal design; the agent reads the simulator through one-way attention and cannot bias it.

06

What is still open

9.6 s of memory, a frozen simulator the policy could exploit, thin statistics on the headline number, and no code or weights.

Questions for discussion

- 01 How much of the diamond result is the world model versus PMPO? Would a stronger offline-RL baseline on world-model features (no imagination) close the gap?
- 02 Is human play-testing on 16 tasks a sufficient measure of 'accurate simulation'? What automated long-horizon metric would you propose?
- 03 The transformer is frozen during RL and the reward head is learned: where would you expect exploitation to appear first, and how would you detect it offline?
- 04 9.6 s of context: which memory mechanism (retrieval, state-space layers, compressed summaries) fits a block-causal diffusion transformer best?
- 05 The recipe needs ~100 h of action labels per domain. What does the path to internet-scale video look like: inverse dynamics models, or Dreamer 4's 'no-action' embedding at scale?
- 06 What would convince you that this transfers to real robots, and what evaluation would you run first?

References & resources

- **Paper:** Hafner*, Yan*, Lillicrap. Training Agents Inside of Scalable World Models. arXiv:2509.24527 (29 Sep 2025). <https://arxiv.org/abs/2509.24527>
- **Project page & videos:** <https://danijar.com/project/dreamer4/> (overview video [youtube.com/watch?v=oDlBtTcX0g0](https://www.youtube.com/watch?v=oDlBtTcX0g0), four uncut 60-min 'Diamond Challenge' evaluation episodes, 51 human-interaction clips)
- **Authors in conversation:** TalkRL E73 'Danijar Hafner on Dreamer v4' (10 Nov 2025), transcript available; Hack Club AMA (Dec 2025)
- **Ingredients:** Chen et al., Diffusion Forcing (2407.01392) · Frans, Hafner, Levine, Abbeel, Shortcut Models (2410.12557) · Chen et al., MAETok (2502.03444) · Baker et al., VPT (2206.11795) · Hafner et al., DreamerV3 (2301.04104)
- **Independent reproductions / follow-ups:** next-state/open-dreamer (JAX, 1.6 B Minecraft world model, Jul 2026) · lamCreateAI/Dreamerv4-MC (weights on HF) · nicklashansen/dreamer4 & MMBench2 (Hansen & Wang, 2606.27326) · lucidrains/dreamer4 (PyTorch)
- **Everything above, archived:** <https://github.com/HarmannBitte/Dreamer-4> (PDFs, repo snapshots, articles, transcripts, the official clips used in this deck, and a MEDIA_INDEX.md)

The test bed: the offline diamond challenge

No environment interaction at all: everything is learned from 2,541 hours of human gameplay video

- **Data:** OpenAI VPT contractor recordings (subsets 6–10), 2,541 h at 360×640, 20 FPS, with mouse & keyboard labels; 90/10 train/eval split by 5-minute chunk
- **Actions:** 23 binary keys + 121-way mouse class (11×11 foveated bins, VPT convention); no abstract 'craft' actions
- **Evaluation:** 1,000 fresh random worlds, empty inventory, 60 min each (72,000 actions at 20 FPS); a fixed prompt sequence walks the agent through the 12-item tech tree
- Why it is hard: diamonds sit ~24,000 actions deep for a human; rewards are sparse; offline data covers the late tech tree thinly; every skill (chop → craft → mine → smelt) must chain without a single trial in the real game
- **Baselines, all on the same data:** VPT fine-tuned, BC, BC without task input, a VLA built on Gemma 3, and 'WM + BC' (Dreamer 4 without imagination RL)



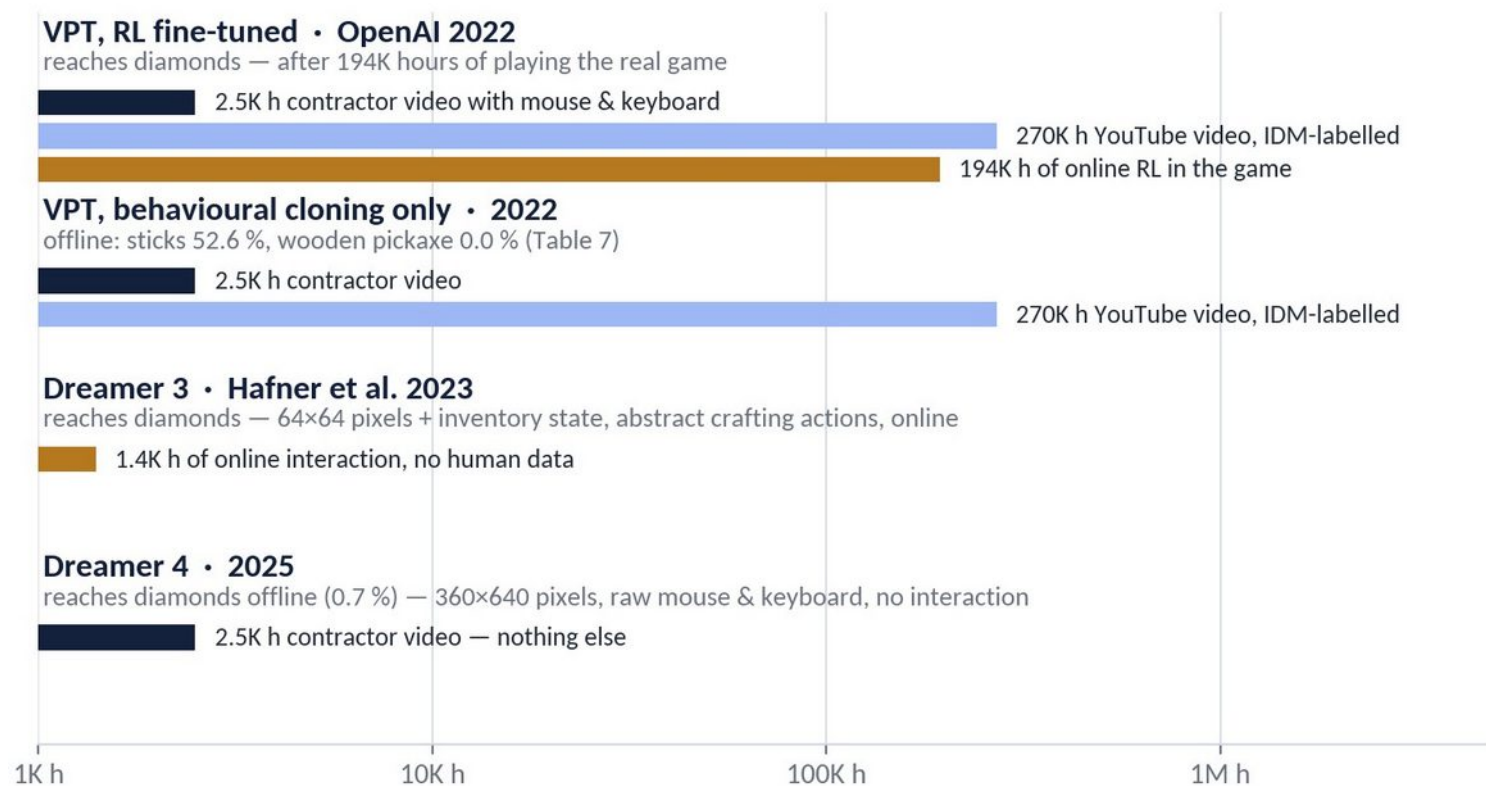
Diamond ore, as simulated by the world model during a human 'mine diamonds' session (project page).

THE 12 MILESTONES (TABLE 5)

log → planks → crafting table → stick → wooden pickaxe → cobblestone
→ stone pickaxe → iron ore → furnace → iron ingot → iron pickaxe →
diamond

Who reached diamonds before, and what it cost

Table 3: same goal, very different regimes. The hours of data and interaction behind each agent



Hours on a log scale (Table 3 of the paper). Dreamer 4 uses the same 2.5K contractor hours that VPT used to train its inverse-dynamics model (IDM) — and nothing else: no pseudo-labelled web video, no environment interaction.

VPT (OPENAI, 2022)

Showed that web video plus an inverse-dynamics model can bootstrap a mouse-and-keyboard policy. Diamonds still took 194K hours of trial and error in the live game, and the pure behavioural-cloning variant never crafts a wooden pickaxe offline.

DREAMER 3 (2023)

Reached diamonds from scratch by online RL inside a small recurrent world model, at 64×64 pixels, with inventory state as input and abstract crafting actions instead of mouse clicks in menus.

DREAMER 4'S BET

Keep the human interface (360×640 pixels, raw mouse & keyboard), drop environment interaction entirely, and get the missing experience from imagination instead of the game, with 100× less data than the keyboard-and-mouse agents before it (Table 3 caption).

Overview: three phases, one transformer

Algorithm 1 in the paper

Phase 1

World-model pretraining

- Train the causal tokenizer (masked autoencoding), then freeze it
- Train the interactive dynamics model on tokenized video $\pm$ actions with shortcut forcing
- Inputs: all 2,541 h, labeled or not

→

Phase 2

Agent fine-tuning

- Insert agent tokens with task embedding; add policy & reward heads
- Behavioural cloning with multi-token prediction ($L = 8$) on task-relevant data
- Dynamics loss continues on uniform data so the model stays honest

→

Phase 3

Imagination training

- Freeze the transformer; train policy + value heads only
- Roll out the policy inside the world model, reward from the learned reward head
- PMPO with reverse-KL to the frozen BC policy; $\gamma = 0.997$

- One block-causal transformer design is used for the tokenizer, the dynamics model and the agent backbone; policy and value are small heads on 'agent tokens'
- Data mix in phases 2 and 3: 50 % uniform sequences, 50 % task-relevant; the BC loss is applied only to relevant data and the dynamics loss only to uniform data ('to avoid optimistic generations')

Algorithm 1 in full (1/2): Phase 1, world-model pretraining

Each step of the recipe with its objective written out; equation numbers are the paper's (§3.1 and §3.2)

STEP 1 • TRAIN THE TOKENIZER ON VIDEOS, EQ. (5)

$$\mathcal{L}_{\text{tok}}(\theta) = \mathcal{L}_{\text{MSE}}(\theta) + 0.2 \mathcal{L}_{\text{LPIPS}}(\theta)$$

- Masked autoencoding: in every frame a random fraction $p \sim \text{U}(0, 0.9)$ of the 16×16 input patches is replaced by a learned mask embedding; the decoder must reconstruct the full frame from the 256 latent tokens (512×16 tanh bottleneck)
- MSE keeps colours and layout; LPIPS keeps textures. Like every loss term in the paper, both are divided by a running estimate of their root-mean-square, which is why a fixed weight of 0.2 works
- Encoder and decoder are causal in time, so the same tokenizer runs frame by frame at inference, with $p = 0$
- ≈ 400 M parameters; trained once on all 2,541 h of video, then frozen for everything that follows

STEP 2 • TRAIN THE WORLD MODEL ON TOKENIZED VIDEO ± ACTIONS, EQS. (6)-(8)

$$z_0 \sim \mathcal{N}(0, I), \quad z_1 \sim \mathcal{D}, \quad (\tau, d) \sim p(\tau, d), \quad \tilde{z} = (1 - \tau) z_0 + \tau z_1, \quad \hat{z}_1 = f_\theta(\tilde{z}, \tau, d, a)$$

Every frame is corrupted to its own signal level τ_{t} ; the network sees the noisy latents, τ , the step size d and the actions, and predicts the clean latents (x-prediction). Loss (7): a flow-matching term at the smallest step size, a bootstrap term for larger steps:

$$\mathcal{L}(\theta) = \|\hat{z}_1 - z_1\|_2^2 \quad \text{if } d = d_{\min}$$

$$\mathcal{L}(\theta) = (1 - \tau)^2 \left\| \frac{\hat{z}_1 - \tilde{z}}{1 - \tau} - \text{sg}\left(\frac{b' + b''}{2}\right) \right\|_2^2 \quad \text{otherwise}$$

$$b' = \frac{f_\theta(\tilde{z}, \tau, d/2, a) - \tilde{z}}{1 - \tau}, \quad z' = \tilde{z} + b' d/2, \quad b'' = \frac{f_\theta(z', \tau + d/2, d/2, a) - z'}{1 - (\tau + d/2)}$$

Every term is multiplied by the ramp weight (8): $w(\tau) = 0.9\tau + 0.1$

- The bootstrap target b' , b'' is two half-steps computed by the model itself (stop-gradient); it is formed in v -space and scaled back to x -space by $(1 - \tau)^2$
- Unlabeled frames use a learned 'no action' embedding, so labeled and unlabeled video train the same loss; at inference $K = 4$ steps of $d = 1/4$ per frame, past frames kept at $\tau_{\text{ctx}} = 0.1$

Algorithm 1 in full (2/2): Phases 2 and 3

Agent fine-tuning with eqs. (7) + (9), then imagination training with eqs. (10) and (11); the Phase 1 transformer stays the backbone (§3.3)

PHASE 2 • AGENT FINE-TUNING, EQS. (7) + (9)

- Insert agent tokens as an extra modality in every frame. They receive the task embedding q_t and attend to latents, actions and the past; nothing attends back to them, so the world model cannot be steered by the task
- Policy and reward heads are small MLPs on the agent-token outputs h_t , with one output layer per prediction distance $n = 0 \dots L$ (multi-token prediction):

$$\mathcal{L}(\theta) = - \sum_{n=0}^L \ln p_{\theta}(a_{t+n} | h_t) - \sum_{n=0}^L \ln p_{\theta}(r_{t+n} | h_t), \quad L = 8$$

- Reward head: symexp twohot distribution over the game-event rewards. Policy head: categorical for the mouse class, vectorised binary for the 23 keys
- The pretraining loss (7) keeps running on the same noised latents, so the world model does not degrade. Data mix: 50 % uniform sequences (dynamics loss) and 50 % task-relevant sequences (BC and reward loss)
- Output of this phase: the 'WM + BC' agent (iron pickaxe 16.9 %, no diamonds)

PHASE 3 • IMAGINATION TRAINING, EQS. (10) + (11)

- Freeze the transformer. Add a value head and keep a frozen copy of the policy head as behavioural prior π_{prior}
- Rollouts start from dataset contexts, one per context: the flow head samples the next latents ($K = 4$ steps), the policy head samples the action; the reward head labels each step r_t , the value head v_t
- Value head: TD learning on λ -returns, $\gamma = 0.997$, $c_t = 1$ for non-terminal states, symexp twohot output

$$\mathcal{L}(\theta) = - \sum_{t=1}^T \ln p_{\theta}(R_t^{\lambda} | s_t), \quad R_t^{\lambda} = r_t + \gamma c_t [(1 - \lambda) v_t + \lambda R_{t+1}^{\lambda}], \quad R_T^{\lambda} = v_T$$

Policy head, PMPO; only the sign of the advantage enters:

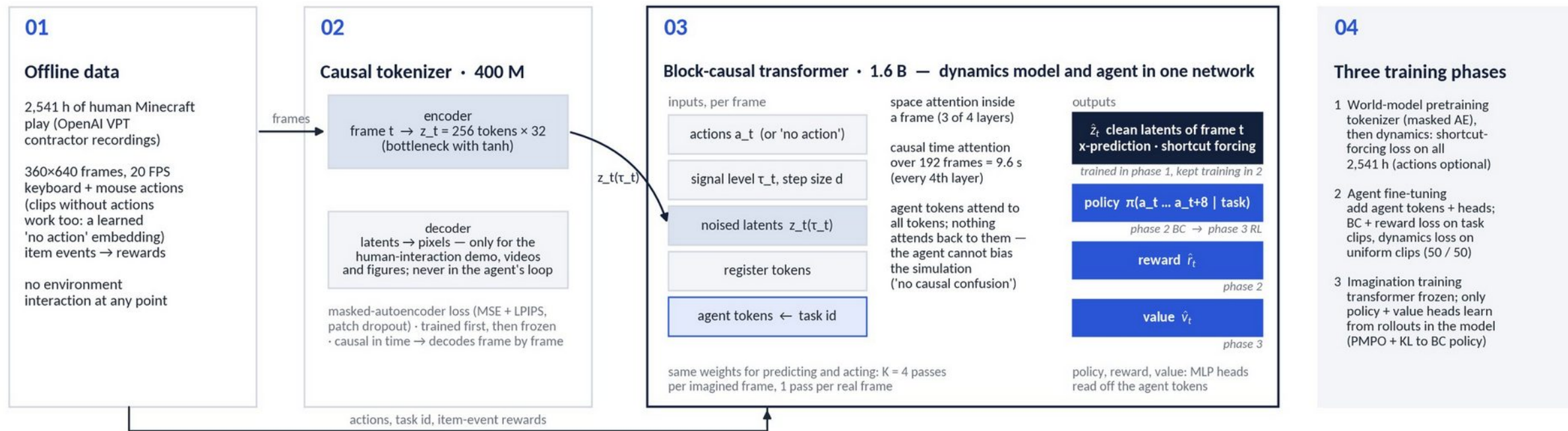
$$\mathcal{L}(\theta) = \frac{1 - \alpha}{|\mathcal{D}^-|} \sum_{i \in \mathcal{D}^-} \ln \pi_{\theta}(a_i | s_i) - \frac{\alpha}{|\mathcal{D}^+|} \sum_{i \in \mathcal{D}^+} \ln \pi_{\theta}(a_i | s_i) + \frac{\beta}{N} \sum_{i=1}^N \text{KL}[\pi_{\theta}(\cdot | s_i) \| \pi_{\text{prior}}]$$

$$A_t = R_t^{\lambda} - v_t, \quad \mathcal{D}^+ = \{s_i : A_i \geq 0\}, \quad \mathcal{D}^- = \{s_i : A_i < 0\}, \quad \alpha = 0.5, \quad \beta = 0.3, \quad \gamma = 0.997$$

- $\alpha = 0.5$ balances the positive and negative sets; the KL runs in the reverse direction of the original PMPO to keep the policy inside the prior's support; all three terms are in nats. Only the policy and value heads receive gradients

How the pieces fit together (1/2): data → tokenizer → one transformer

What flows between the components, and which phase trains which output



- Two block-causal transformers: the tokenizer (400 M) turns each frame into 256 latent tokens; the dynamics/agent transformer (1.6 B) predicts the clean next latents and reads policy, reward and value off its agent tokens (MLP heads, multi-token prediction $L = 8$)
- The decoder sits outside the agent's loop: pretraining, fine-tuning and imagination all happen in latent space, and pixels are only rendered for humans
- Agent tokens have one-way attention: they see everything and nothing sees them, so adding the agent cannot change what the simulator predicts

How the pieces fit together (2/2): the two loops

Training inside the model, and acting in the real game

05

Acting in the real game (evaluation)

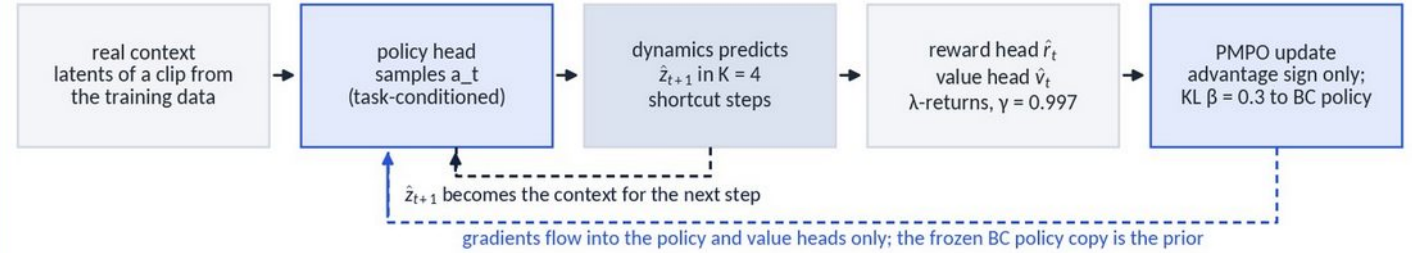


The world model generates nothing at test time — it supplies the representation the policy head reads. Only encoder + transformer + policy head run, one pass per frame.

1,000 episodes × 60 min, fresh random worlds, empty inventory, a fixed prompt sequence of tasks (log → planks → ... → iron pickaxe → diamond). Success = the item shows up in the inventory.

06

Phase 3 • the imagination loop — inside the frozen model, entirely in latent space

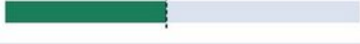
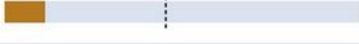
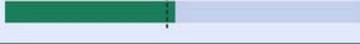


Each rollout starts from real frames, then follows the current policy through the model's imagination, one rollout per context. Rewards come from the learned reward head (never from the game); the reverse KL keeps the policy in the space of behaviours seen in the data. Result on the offline diamond challenge: BC on the same representations reaches the iron pickaxe in 16.9 % of episodes and never a diamond; after imagination training it is 29 % and 0.7 %.

- Imagination loop (phase 3): start from real latents → policy head samples an action → dynamics predicts the next latents in $K = 4$ shortcut steps → reward and value heads score the step → PMPO with reverse KL updates the policy and value heads only; one rollout per context, transformer frozen
- Acting (evaluation): encoder → transformer → policy head, once per frame at 20 FPS; at test time the world model only provides the representation and generates nothing
- What the loop buys: on the same representations, behavioural cloning reaches the iron pickaxe in 16.9 % of episodes and never a diamond; after imagination training, 29 % and 0.7 % (Table 7)

Minecraft world models, stack by stack (2/2): how they run

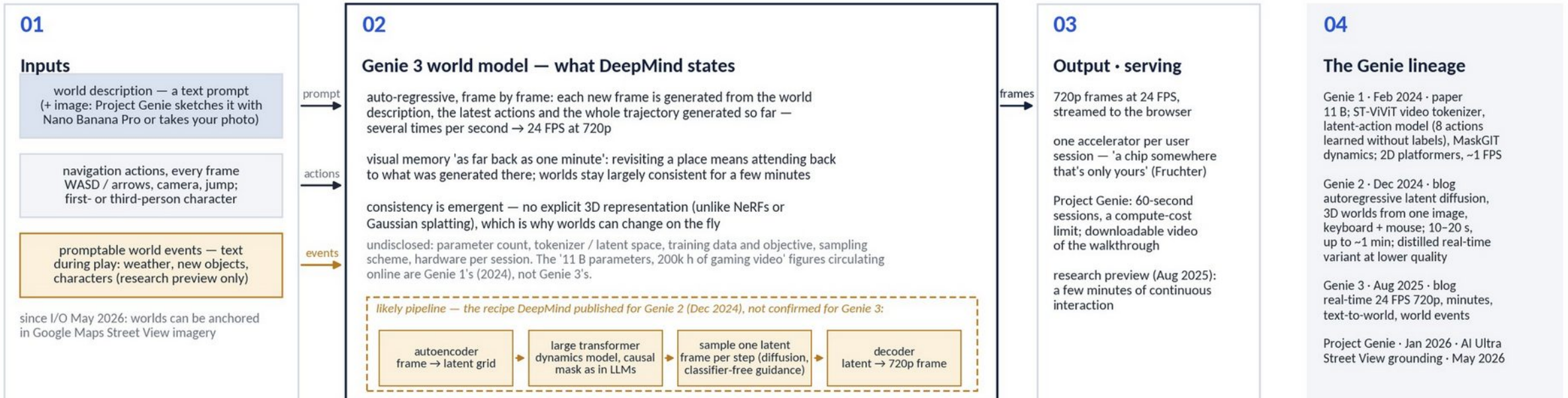
Sampling, context, speed on one H100, the 16-task human play-test, and what each model is used for

layer of the stack Minecraft world models, 2024-25	MineWorld discrete autoregressive	Lucid-v1 latent diffusion / flow	Oasis (small) latent diffusion	Oasis (large) latent diffusion	Dreamer 4 latent flow · shortcut forcing
generating a frame	autoregressive over 336 codes; parallel 'diagonal' decoding of spatially adjacent codes (> 3× faster) — needs the actions in advance → no true real-time play	few flow steps per frame with a KV cache; 20-30 FPS on one RTX 4090 in the released demo	DDIM, 10 steps by default; dynamic noising: noise the context in early passes, remove it later, to curb error build-up	same sampler on Decart's proprietary inference stack, served from several H100s	K = 4 shortcut steps per frame (one forward pass each, step sizes learned by self-distillation); context kept slightly noised
context	0.8 s 	1.0 s 	1.6 s 	1.6 s 	9.6 s = 192 frames at 20 FPS 
speed on one H100 · resolution	2 FPS · 384×224  20 FPS = game rate	44 FPS · 640×360 	20 FPS · 640×360 	~5 FPS · 360×360 	21 FPS · 640×360 
human play-test, 16 tasks	— not evaluable: needs actions ahead of time	0 / 16 tasks generations diverge, interactions fail 	0 / 16 tasks visual degradation, hallucinated structures 	5 / 16 tasks 	14 / 16 tasks 
beyond the simulator	same network can act as a policy (predict action tokens); no RL	simulator only	simulator only (tech demo)	simulator only (demo / product)	agent tokens + policy / reward / value heads; RL in imagination → diamonds from offline data only

- Speed alone does not explain the results: Lucid-v1 is the fastest model in the table and solves none of the tasks; MineWorld cannot be played at all because its parallel decoding needs the actions in advance
- Context length does: 9.6 s versus 0.8–1.6 s is what lets the player look away and back, and the two tasks Dreamer 4 still fails are the ones that exceed that window
- Only Dreamer 4 is used as a training ground: the others are simulators or demos; here the same network also hosts the policy trained inside it

Genie 3 (1/2): what is public

Google DeepMind, Aug 2025; blog post and model page only. Plain boxes = stated; amber = inferred from the Genie 2 recipe; grey = undisclosed



- A different problem: any text-prompted world, 24 FPS at 720p, about a minute of visual memory, navigation-level actions, versus one game, full mouse and keyboard, 9.6 s of context at 21 FPS and 360p
- Why the paper cannot compare (§4.2): Genie 3 exposes only camera actions and one generic 'interact' button, so the 16 Minecraft tasks are not expressible; nothing about its size, data or objective is published, and the '11 B / 200k hours' figures that circulate are Genie 1's

Dreamer 3 vs Dreamer 4

Appendix F: what changed, and what the higher-fidelity model costs

	Dreamer 3 (Nature 2025)	Dreamer 4
World model	RSSM (recurrent + variational latent)	Block-causal diffusion/flow transformer, shortcut forcing
Minecraft input	64×64 pixels + inventory state; abstract crafting actions	360×640 pixels only; raw mouse & keyboard
Data	1.4 K h of <i>online</i> interaction, no human data	2.5 K h of <i>offline</i> human data, no interaction
RL	Return normalisation + entropy bonus	PMPO (advantage sign) + KL to BC prior
Speed	~1,000× faster than a diffusion transformer	~30× faster than diffusion forcing; ~40× slower than the RSSM

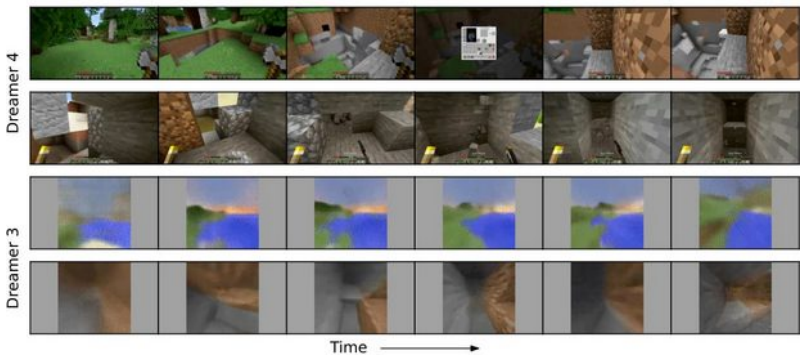


Figure 11: multi-step generations, Dreamer 4 (top) vs Dreamer 3 (bottom).



Generation speed on a log scale (paper, TalkRL): ~30× faster than diffusion forcing, ~40× slower than the RSSM.

Limitations the paper itself states

From the discussion section and the authors' talks

- 01 **Diamonds are rare:** 0.7 % success; the authors call the pipeline a 'reliable and performant starting point', not a solved task
- 02 **Memory:** 9.6 s of context; walk into a house for 30 s and the outside is regenerated. Long-term memory is the first item on the authors' roadmap
- 03 **Inventory drift:** small UI elements and item counts blur over long horizons; small objects and long-range semantic correlations are what video models learn last
- 04 **Frozen, exploitable simulator:** no epistemic-uncertainty estimate; the policy can exploit model errors (invalid crafting), currently mitigated only by the KL constraint
- 05 **Two-stage training:** tokenizer is trained separately and frozen; the authors expect end-to-end training eventually
- 06 **Single domain for the agent:** RL results exist only for Minecraft; robotics and kitchen video are qualitative world-model demos
- 07 **Compute:** hundreds of TPU v5p chips per run; not a recipe individuals can reproduce at full scale

Appendix A: Configuration and hyper-parameters

Component	Setting
Tokenizer	Masked autoencoder; 16×16 patches; patch dropout $\sim U(0, 0.9)$; loss MSE + 0.2·LPIPS; bottleneck 512×16 with tanh $\rightarrow N_z = 256 \text{ tokens} \times 32 \text{ dims}$; causal in time; $\approx 400 \text{ M params}$; frozen after pretraining
Dynamics model	$\approx 1.6 \text{ B params}$; pre-RMSNorm, RoPE, SwiGLU, QK-norm, logit soft-capping, register tokens; space attention in 3/4 layers, causal time attention every 4th layer; GQA; context C = 192 frames (9.6 s); alternating batch lengths T1/T2 = 64/256
Shortcut forcing	per-frame signal level τ , step size d; x-prediction; bootstrap target formed in v-space, loss scaled to x-space by $(1-\tau)^2$; ramp weight $w(\tau) = 0.9\tau + 0.1$; K = 4 sampling steps; context noise $\tau_{\text{ctx}} \approx 0.1$
Actions	23 binary keys + 121-way mouse (μ -law, 11×11 foveated bins, VPT convention); learned 'no action' embedding for unlabeled video
Agent (phase 2)	agent tokens with one-hot task (20 tasks), one-way attention; BC with multi-token prediction L = 8; task-conditioned scalar reward head; data mix 50 % uniform / 50 % task-relevant
Imagination RL (phase 3)	transformer frozen; policy + TD value heads; PMPO with advantage sign, $\alpha = 0.5$, reverse-KL $\beta = 0.3$ to frozen BC policy; $\gamma = 0.997$
Evaluation	1,000 episodes $\times$ 60 min, empty inventory, random worlds, fixed prompt sequence; human interaction: 16 tasks on one H100 at 21 FPS
Compute / data	256–1,024 TPU v5p; 2,541 h VPT contractor video (subsets 6–10), 90/10 split by 5-min chunk; ablations: 48 h runs, FVD on 1,024 $\times$ 384-frame generations

Appendix C: Full success-rate table (Table 7, %)

Milestone	VPT (ft)	BC (no task)	BC	VLA (Gemma 3)	WM + BC	Dreamer 4
Log	84.3	71.4	97.3	98.5	99.6	99.1
Planks	65.3	68.6	95.7	98.3	99.6	98.9
Crafting table	4.7	63.8	93.5	97.2	99.1	98.5
Stick	52.6	62.4	95.0	97.7	98.9	98.7
Wooden pickaxe	0.0	33.8	86.5	94.1	97.3	96.6
Cobblestone	6.9	32.0	83.9	91.6	97.2	95.9
Stone pickaxe	0.0	8.8	53.8	76.7	89.4	90.1
Iron ore	0.1	3.6	26.5	46.3	62.9	66.7
Furnace	0.0	4.0	16.2	42.4	51.1	58.1
Iron ingot	0.1	0.2	4.3	22.5	27.8	39.5
Iron pickaxe	0.0	0.0	0.6	11.2	16.9	29.0
Diamond	0.0	0.0	0.0	0.0	0.0	0.7

1,000 evaluation episodes of 60 minutes each; all agents trained on the same 2,541 h of contractor data (VPT additionally pretrained on 270 K h of web video).

Appendix D: Glossary

Term	Meaning in this paper
World model	Learned model that predicts future observations (here: latent frames) given past observations and actions
Imagination training	Training the policy on rollouts generated by the world model instead of the real environment
Tokenizer / latents	Encoder-decoder mapping each frame to 256 continuous tokens the dynamics model operates on
Block-causal attention	Full attention within a frame, causal across frames; enables frame-by-frame online generation
Diffusion forcing	Diffusion training with an independent noise level per frame in a sequence
Shortcut model	Diffusion/flow model conditioned on step size, trained to be self-consistent across step sizes, which allows few-step sampling
x- vs v-prediction	Network predicts the clean latent (x) vs the velocity $x - \epsilon(v)$; Dreamer 4 uses x-prediction
FVD / PSNR / SSIM	Fréchet Video Distance (distribution-level video quality, lower is better) / per-frame reconstruction fidelity metrics
GQA, RoPE, registers	Grouped-query attention (fewer KV heads), rotary position embeddings, extra learnable tokens that absorb attention mass
BC, VLA, PMPO	Behavioural cloning; vision-language-action model (here built on Gemma 3); the sign-based, KL-regularised policy-gradient update used for RL